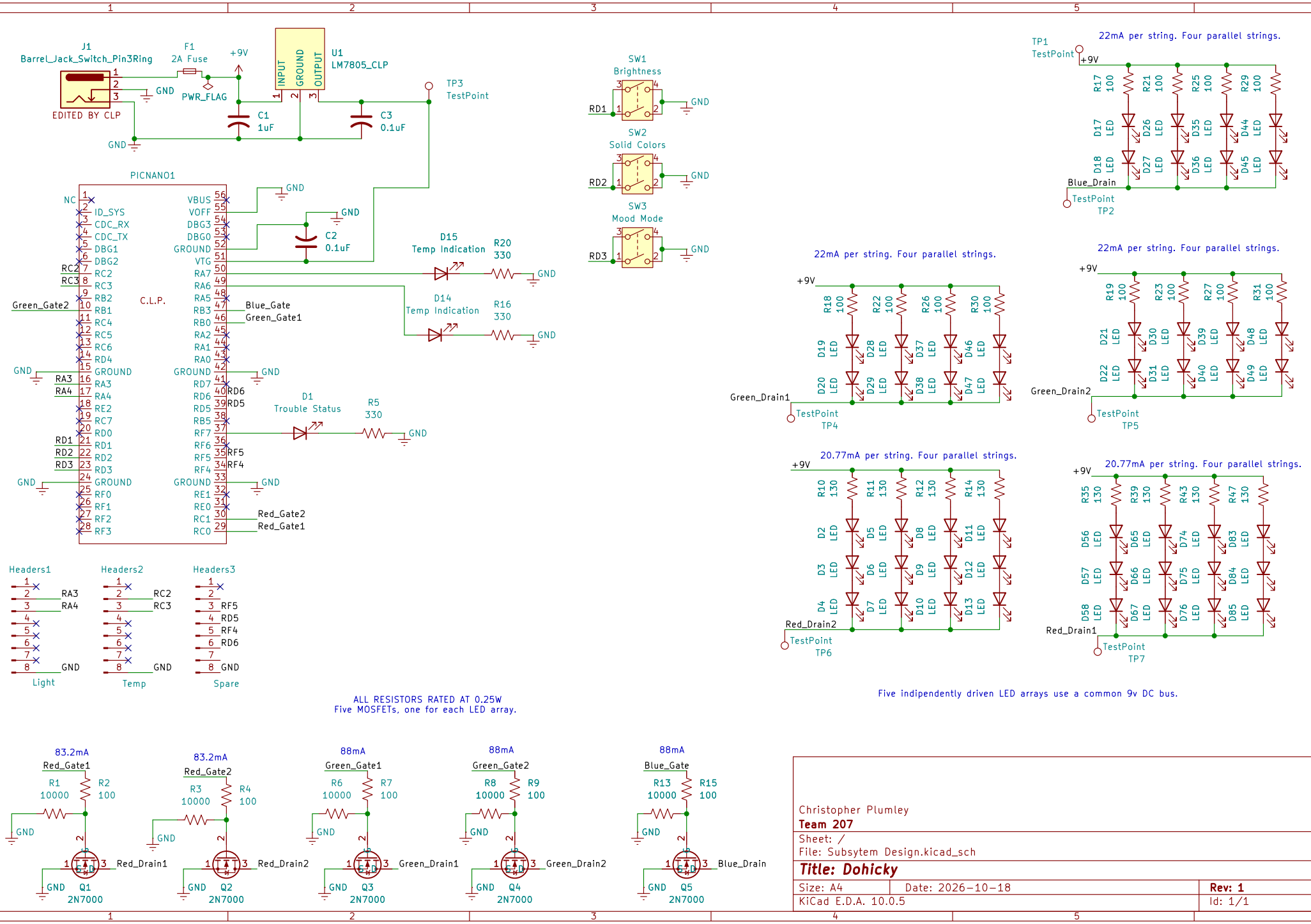

J1
Barrel_Jack_Switch_Pin3Ring
EDITED BY CLP
F1
2A Fuse
PWR_FLAG
+9V
GND
U1
LM7805_CLP
INPUT
GROUND
OUTPUT
C1
1uF
C3
0.1uF
TP3
TestPoint
PICNANO1
C.L.P.
NC
ID_SYS
CDC_RX
CDC_TX
DBG1
DBG2
RC2
RC3
RB2
RB1
RC4
RC5
RC6
RD4
GROUND
RA3
RA4
RE2
RC7
RD0
RD1
RD2
RD3
GROUND
RF0
RF1
RF2
RF3
VBUS
VOFF
DBG3
DBG0
GROUND
VTG
RA7
RA6
RA5
RB3
RB0
RA2
RA1
RA0
GROUND
RD7
RD6
RD5
RB5
RF7
RF6
RF5
RF4
GROUND
RE1
RE0
RC1
RC0
C2
0.1uF
D15
Temp Indication
R20
330
D14
Temp Indication
R16
330
D1
Trouble Status
R5
330
Green_Gate2
Blue_Gate
Green_Gate1
Red_Gate2
Red_Gate1
SW1
Brightness
SW2
Solid Colors
SW3
Mood Mode
RD1
RD2
RD3
Headers1
Light
Headers2
Temp
Headers3
Spare
ALL RESISTORS RATED AT 0.25W
Five MOSFETs, one for each LED array.
83.2mA
Red_Gate1
R1
10000
R2
100
Q1
2N7000
Red_Drain1
83.2mA
Red_Gate2
R3
10000
R4
100
Q2
2N7000
Red_Drain2
88mA
Green_Gate1
R6
10000
R7
100
Q3
2N7000
Green_Drain1
88mA
Green_Gate2
R8
10000
R9
100
Q4
2N7000
Green_Drain2
88mA
Blue_Gate
R13
10000
R15
100
Q5
2N7000
Blue_Drain
22mA per string. Four parallel strings.
TP1
TestPoint
R17 R21 R25 R29
D17 D26 D35 D44
D18 D27 D36 D45
Blue_Drain
TP2
22mA per string. Four parallel strings.
R18 R22 R26 R30
D19 D28 D37 D46
D20 D29 D38 D47
Green_Drain1
TP4
22mA per string. Four parallel strings.
R19 R23 R27 R31
D21 D30 D39 D48
D22 D31 D40 D49
Green_Drain2
TP5
20.77mA per string. Four parallel strings.
R10 R11 R12 R14
D2 D5 D8 D11
D3 D6 D9 D12
D4 D7 D10 D13
Red_Drain2
TP6
20.77mA per string. Four parallel strings.
R35 R39 R43 R47
D56 D65 D74 D83
D57 D66 D75 D84
D58 D67 D76 D85
Red_Drain1
TP7
Five indipendently driven LED arrays use a common 9v DC bus.
Christopher Plumley
Team 207
Sheet: /
File: Subsytem Design.kicad_sch
Title: Dohicky
Size: A4
Date: 2026-10-18
Rev: 1
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